

Automated Employee Onboarding & Offboarding System Using ServiceNow

1. Introduction

1.1 Project Overview

The **Automated Employee Onboarding & Offboarding System** is a ServiceNow-based workflow automation solution designed to streamline the complete employee lifecycle. Traditional onboarding and offboarding processes often involve manual coordination among HR, Managers, IT, Facilities, and Security departments, leading to delays, communication gaps, and reduced operational efficiency.

This project automates the entire process using **Service Catalog, Flow Designer, Custom Tables, Approval Workflows, Catalog Tasks, Service Level Agreements (SLAs), Role-Based Access Controls (ACLs), Reports, and Dashboards**. The system ensures that every employee request is processed through a standardized workflow, improving transparency, accountability, and compliance.

1.2 Purpose

The primary purpose of this project is to replace manual employee onboarding and offboarding activities with an automated digital workflow that improves efficiency, reduces processing time, and minimizes human errors.

The objectives include:

- Automate employee onboarding and offboarding requests.
- Eliminate manual task assignments.
- Route requests for manager approval automatically.
- Generate IT, Facilities, and Security tasks automatically.
- Monitor task completion using SLAs.
- Improve collaboration among departments.
- Provide HR with real-time dashboards and reports.
- Maintain secure access through role-based permissions.

2. Ideation Phase

The Ideation Phase focused on identifying the key challenges faced during employee onboarding and offboarding in organizations. Discussions were conducted to understand the difficulties experienced by HR, Managers, IT, Facilities, and Security teams in managing employee lifecycle activities manually. Various ideas were evaluated, and the most feasible solution was selected based on automation capability, scalability, business value, and ease of implementation.

Through customer problem analysis, empathy mapping, brainstorming, and idea prioritization, the project team identified that ServiceNow provides an ideal platform for automating employee lifecycle processes. The selected solution aims to improve efficiency, reduce manual effort, enhance transparency, and ensure timely completion of onboarding and offboarding activities through workflow automation.

The Ideation Phase includes:

- Problem Statement
- Empathy Map Canvas
- Brainstorming & Idea Prioritization

3. Requirement Analysis

The Requirement Analysis phase involved gathering and analyzing the functional and non-functional requirements of the proposed Employee Onboarding & Offboarding System. The objective was to understand business needs, stakeholder expectations, and system requirements to ensure that the solution addresses real organizational challenges.

During this phase, customer interactions, workflow analysis, and business processes were studied to define the complete employee lifecycle. A customer journey map was prepared to visualize the user experience, while solution requirements were identified to define the core functionalities and quality attributes of the system. The overall analysis served as the foundation for designing and implementing an automated workflow using the ServiceNow platform.

The Requirement Analysis phase includes:

- Customer Journey Map
- Solution Requirements

4. Project Design

4.1 Problem–Solution Fit

The proposed solution addresses delays and inefficiencies in manual employee lifecycle management by introducing workflow automation. ServiceNow automates request submission, approvals, task creation, tracking, and reporting, ensuring consistent execution across departments.

4.2 Proposed Solution

The solution uses the ServiceNow platform to automate employee onboarding and offboarding. HR submits requests through the Service Catalog, which are stored in a custom Employee Lifecycle Management table. Flow Designer routes the request for manager approval and automatically generates tasks for IT, Facilities, and Security. SLA monitoring, notifications, reports, and dashboards provide complete visibility into the process while ACLs ensure secure access to employee data.

4.3 Solution Architecture

Components

- Service Portal
- Service Catalog
- Employee Lifecycle Management Custom Table
- Flow Designer
- Manager Approval
- IT Team
- Facilities Team
- Security Team
- SLA
- Reports
- Dashboard

5. Project Planning & Scheduling

The development was completed using the Agile methodology, structured into three main sprints. The project was broken down into topics and user stories, totaling 51 story points with an average velocity of 17 points per sprint.

Sprint Breakdown

Sprint 1: Roles, Security & Data Architecture (14 Points)

- **Roles & Security:** Created roles for HR, IT, Facilities, and Security; created groups and assigned users; configured ACLs.
- **Data Architecture:** Created the custom Employee Lifecycle Management table; configured fields and form layouts.

Sprint 2: Service Catalog & Flow Designer (18 Points)

- **Service Catalog:** Developed Onboarding and Offboarding Catalog Items; configured variables and UI policies.
- **Flow Designer:** Created the Employee Lifecycle Flow and configured the manager approval process.

Sprint 3: Task Automation, Monitoring & Reporting (19 Points)

- **Task Automation:** Automated creation of IT, Facilities, and Security catalog tasks.
- **Monitoring & Reporting:** Configured SLAs, created reports, designed the HR Dashboard, configured notifications, and performed end-to-end testing.

6. Functional & Performance Testing

Functional Testing

The following functionalities were successfully validated:

- Employee request creation
- Catalog variable validation
- Flow execution
- Manager approval
- IT task generation
- Facilities task generation
- Security task generation
- SLA tracking
- Reports
- Dashboard

Performance Testing

The application was tested for normal operational usage.

Observations:

- Catalog submission completed successfully.
- Flow execution performed correctly.
- Tasks were generated automatically.
- Reports and dashboards updated successfully.
- No significant delays were observed during workflow execution.

7. Results

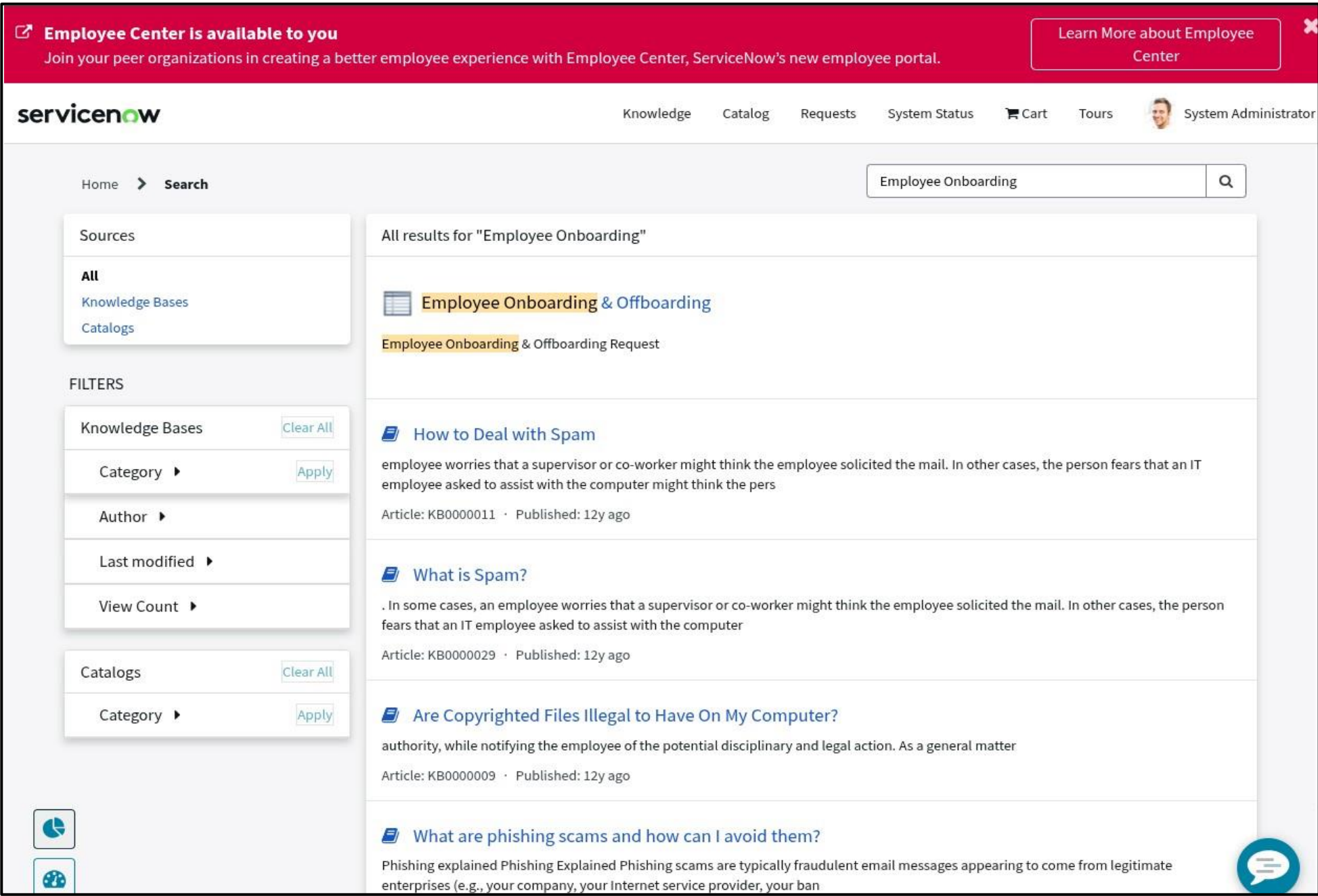
The developed solutions successfully automated the employee onboarding and offboarding process.

Key achievements:

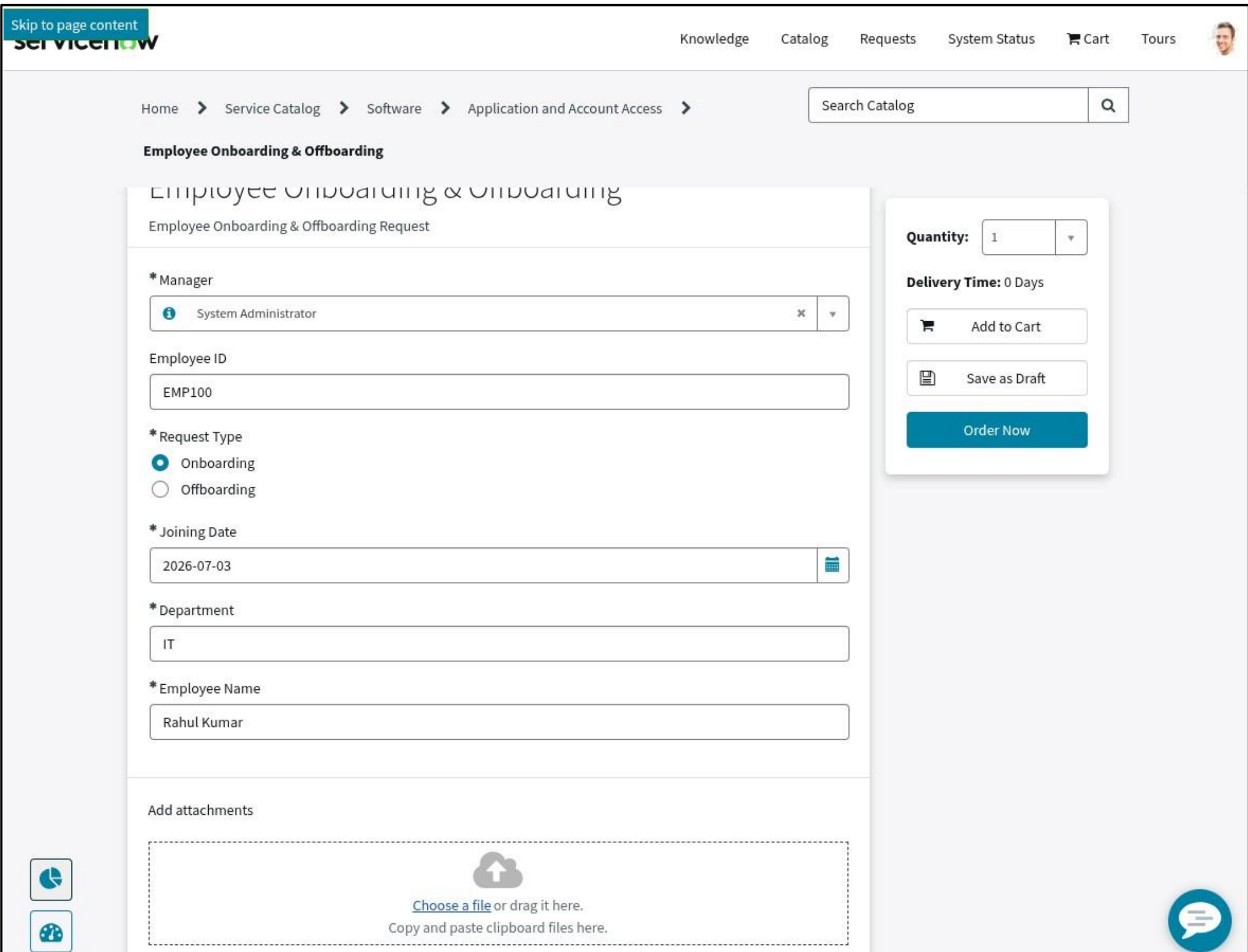
- Automated Service Catalog requests.
- Manager approval workflow.
- Automatic departmental task generation.
- Employee Lifecycle Management records.
- SLA tracking.
- Reports and dashboards.
- Secure access using ACLs.

7.1 Output Screenshots

1.Service Portal



2.EmployeeOnboardingform



3.Request Summary

servicenow

KnowledgeCatalogRequestsSystem StatusCartTours

HomeRequest Summary - REQ0010006

Search Catalog

Submitted :2026-07-02 04:26:49
Request Number : **REQ0010006**
Estimated Delivery : 2026-07-02

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Employee Onboarding & Offboarding	2026-07-02		---	1	---
					Total: \$0.00

4.EmployeeLifecycle Table

ServiceNow DevHome | StudioClassic | UnifiedRequest SummarySowmya1321/E TASK0021061

dev275085.service-now.com/x_2065742_employ_0_request.do?sys_id=bf4d270ec37dc754bcf0b1fdd40131f6&sysparm_record_target=...School

Employee Lifecycle Management Request
TASK0021061

FollowUpdateDelete

NumberTASK0021061

Opened2026-07-02 04:26:51

Assignment group

Opened by

Assigned to

ApprovalRequested

StateOpen

Priority4 - Low

Short description

Description

Employee IDEMP100

Employee NameRahul Kumar

DepartmentIT

ManagerSystem Administrator

Joining Date2026-07-03

Notes

Additional comments (Customer visible)

Work notesPost

Activities: 1System

Field changes • 2026-07-02 04:26:51

No templates are availableCreate A New One?

5.ApprovalList

dev275085.service-now.com/sysapproval_approver_list.do

Approvals Created Search Actions on selected rows...

All

State	Approver	Comments	Approval for	Created
Search	Search	Search	Search	Search
Approved	System Administrator		TASK0021061	2026-07-02 04:26:51
Approved	System Administrator		TASK0021038	2026-07-01 09:25:22
Approved	System Administrator		TASK0021037	2026-07-01 09:24:41
Rejected	Fred Luddy		(empty)	2026-02-02 11:19:33
Requested	Fred Luddy		(empty)	2026-02-02 11:17:03
Requested	Fred Luddy		(empty)	2026-02-02 11:15:44
Approved	Eric Schroeder		CHG0000009	2026-02-01 22:31:51
Approved	Fred Luddy		CHG0000009	2026-02-01 22:31:51
Approved	Bow Ruggeri		CHG0000009	2026-02-01 22:31:50
Approved	David Loo		CHG0000009	2026-02-01 22:31:50
Requested	cab approver		CHG0000079	2024-08-30 06:35:36
Requested	Bernard Laboy		CHG0000079	2024-08-30 06:35:36
Requested	Luke Wilson		CHG0000079	2024-08-30 06:35:35
Requested	Howard Johnson		CHG0000079	2024-08-30 06:35:35
Requested	Christen Mitchell		CHG0000079	2024-08-30 06:35:35
Requested	Ron Kettering		CHG0000079	2024-08-30 06:35:35
Requested	Ron Kettering		CHG0000059	2024-08-30 06:35:27
Requested	Bernard Laboy		CHG0000059	2024-08-30 06:35:27
Requested	Luke Wilson		CHG0000059	2024-08-30 06:35:27
Requested	cab approver		CHG0000059	2024-08-30 06:35:27

6.CatalogTasks

dev275085.service-now.com/sc_task_list.do

Catalog Tasks Number Search Actions on selected rows... New

All

Number	Priority	State	Short description	Assignment group	Assigned to
Search	Search	Search	Search	Search	Search
SCTASK0010020	4 - Low	Open	Provide building and system access	Security Team	(empty)
SCTASK0010019	4 - Low	Open	Prepare workstation and ID card	Facilities Team	(empty)
SCTASK0010018	4 - Low	Open	Create IT account for employee	IT Team	(empty)
SCTASK0010017	4 - Low	Open	Provide building and system access	Security Team	(empty)
SCTASK0010016	4 - Low	Open	Prepare workstation and ID card	Facilities Team	(empty)
SCTASK0010015	4 - Low	Open	Create IT account for employee	IT Team	(empty)
SCTASK0010014	4 - Low	Open	Provide building and system access	Security Team	(empty)
SCTASK0010013	4 - Low	Open	Prepare workstation and ID card	Facilities Team	(empty)
SCTASK0010012	4 - Low	Open	Create IT account for employee	IT Team	(empty)

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7.Flow Designer

Workflow Studio | Employee Onboarding & Offboarding... | Active | View: [Diagram] [Code] | Test | Edit flow | ...

TRIGGER

- Service Catalog

ACTIONS

- Get Catalog Variables from Employee Onboarding & Offboarding
- Create Employee Lifecycle Management Request Record
- Ask For Approval on Employee Lifecycle Management Request
- If Manager Approval Check**
 - then
 - Create Catalog Task Record
 - Create Catalog Task Record
 - Create Catalog Task Record
 - Update Employee Lifecycle Management Request Record

ERROR HANDLER: [Toggle]

Read-only | Status: Published | Application: Employee Lifecycle Management

Data Expand All

- Flow Variables
- Trigger - Service Catalog
- 1 - Get Catalog Variables
- 2 - Create Record
- 3 - Ask For Approval
- 4 - If
- 5 - Create Record
- 6 - Create Record
- 7 - Create Record
- 8 - Update Record

8. ServiceCatalog

dev275085.service-now.com/sc_cat_item.do?sys_id=4c74a67dc3754754bcf0b1fdd4013106&sysparm_record_target=sc_cat_item&sysp...

Catalog Item | Employee Onboarding & Offboarding | Copy | Try It | Update | Edit in Catalog Builder | Delete

Build and modify items faster with the improved [Catalog Builder](#).

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Employee Onboarding & Offboarding | Application: Employee Lifecycle Management | Active: ☒ | Fulfillment automation level: Unspecified

Catalogs: Service Catalog | Category: Application and Account Access | State: -- None -- | Checked out: -- None -- | Owner: System Administrator

Item Details | Process Engine | Picture | Pricing | Portal Settings

Short description: Employee Onboarding & Offboarding Request

Description: [Rich Text Editor]

No templates are available | Create A New One?

dev275085.service-now.com/sc_cat_item.do?sys_id=4c74a67dc3754754bcf0b1fdd4013106&sysparm_record_target=sc_cat_item&sysp...
 School

Catalog Item
Employee Onboarding & Offboarding
Copy Try It Update Edit in Catalog Builder Delete

Press Alt+0 for help

Add relevant tags to the Meta field using comma-separated list of tags. These tags will be used while searching the item. Not applicable if AI Search is configured.

Meta

Copy Try It Update Edit in Catalog Builder Delete

Related Links
[Item Diagnostic](#)
[Run Point Scan](#)

Variables (7) | Variable Sets | Catalog UI Policies (2) | Catalog Client Scripts | Available For | Not Available For | Categories (1) | Catalogs (1) | Catalog Data Lookup Definitions

Related Articles | Related Catalog Items | Assigned Topics

Order Search

Actions on selected rows... New

Catalog item = Employee Onboarding & Offboarding

Type	Question	Order
Reference	Manager	
Single Line Text	Employee ID	
Multiple Choice	Request Type	
Date	Exit Date	
Date	Joining Date	
Single Line Text	Department	
Single Line Text	Employee Name	

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No templates are available [Create A New One?](#)

9.CustomTable

dev275085.service-now.com/sys_db_object.do?sys_id=88351a7dc3f14754bcf0b1fdd4013146&sysparm_record_target=sys_db_object&...
 School

Table
Employee Lifecycle Management Request
Delete Update Delete All Records

Extends table **Request Task**

Columns | Controls | Application Access

Table Columns Display Search

1 to 20 of 81 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Joining Date	Date	(empty)	40		false
Employee ID	String	(empty)	40		false
Request Type	Choice	(empty)	40		false
Task type	System Class Name	(empty)	80	javascript:current.getTableName();	false
Status	Choice	(empty)	40		false
Workflow activity	Reference	Workflow Activity	32		false
Domain Path	Domain Path	(empty)	255	/	false
Contract	Reference	Contract	32		false
Assignment group	Reference	Group	32		false
Approval	String	(empty)	40	not requested	false
Duration	Duration	(empty)	40		false
Knowledge	True/False	(empty)	40	false	false
Correlation display	String	(empty)	100		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Short description	String	(empty)	160		false
Company	Reference	Company	32		false
Approval set	Date/Time	(empty)	40		false
Opened by	Reference	User	32	javascript:gs.getUserID()	false

No templates are available [Create A New One?](#)

10.SLA

dev275085.service-now.com/contract_sla.do?sys_id=edb93a79c3394754bcf0b1fdd4013135&sysparm_record_target=contract_sla&sys...

SLA Definition
Employee Onboarding SLA

An SLA starting now will breach on 2026-07-04 04:37:05 (Actual elapsed time: 2 Days)

Name:

Type:

Target:

Table:

Flow:

Enable logging: ☐

Active: ☒

Application:

Duration type:

* Duration: Days Hours

Schedule source:

* Schedule:

Timezone source:

Start condition:

Pause condition:

Stop condition:

Reset condition:

Start condition: is

Retroactive start: ☐

When to cancel:

Cancel condition: is

Update Delete

Related Links

Show Schedule

No templates are available [Create A New One?](#)

dev275085.service-now.com/task_sla_list.do

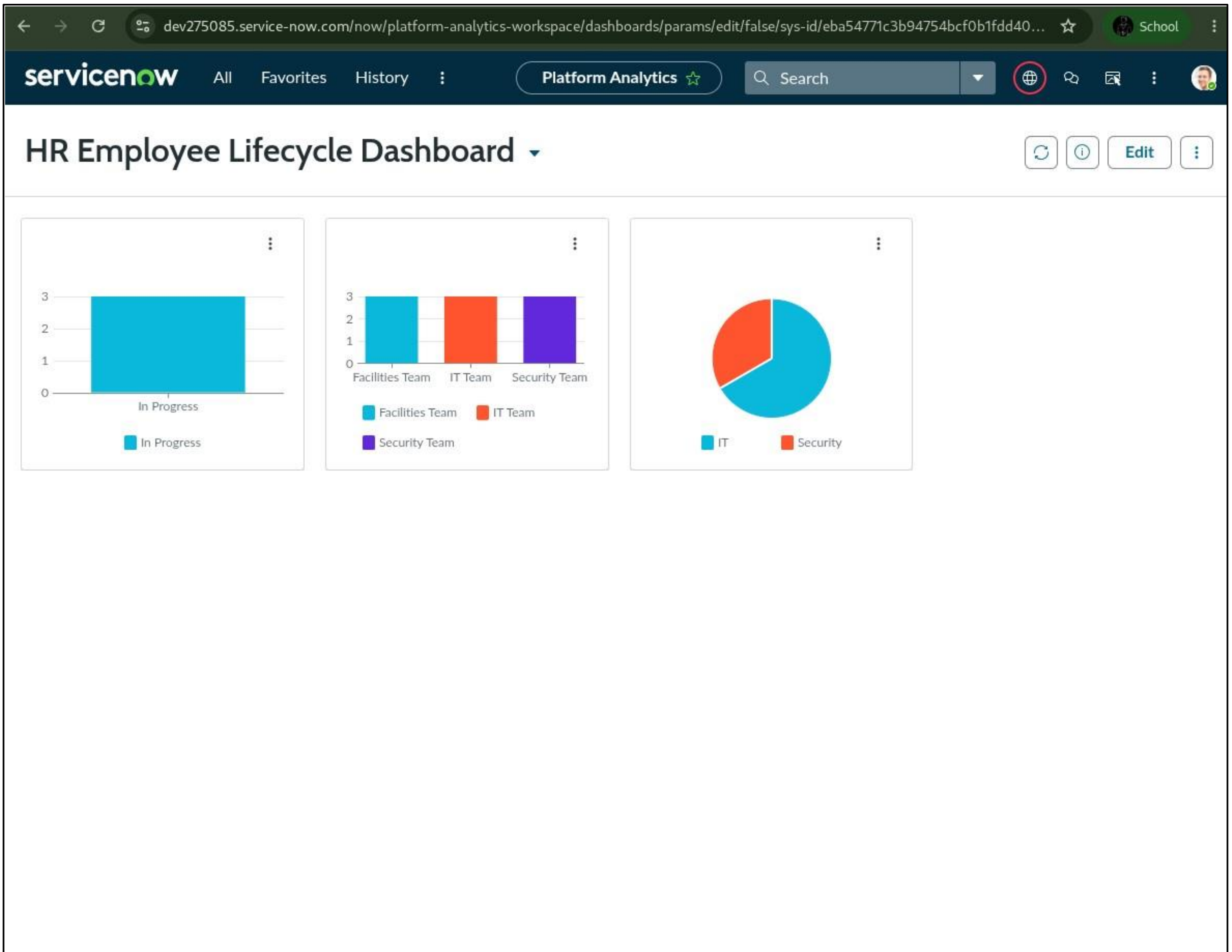
Task SLAs Task Search

Actions on selected rows... New

Task	SLA definition	Type	Target	Stage	Business time left	Business elapsed time	Business elapsed percentage
SCTASK0010014	Employee Onboarding SLA	SLA		In progress	1 Day 12 Hours 34 Minutes	11 Hours 25 Minutes	23.81%
INC0000050	Priority 1 resolution (1 hour)	SLA	Resolution	In progress	0 Seconds	22 Days 19 Hours 4 Minutes	54,707.89%
SCTASK0010012	Employee Onboarding SLA	SLA		In progress	1 Day 12 Hours 34 Minutes	11 Hours 25 Minutes	23.81%
SCTASK0010013	Employee Onboarding SLA	SLA		In progress	1 Day 12 Hours 34 Minutes	11 Hours 25 Minutes	23.81%
INC0000060	Priority 1 resolution (1 hour)	SLA	Resolution	Cancelled	0 Seconds	7 Hours 3 Minutes	705.17%
INC0000027	Priority 2 resolution (8 hour)	SLA	Resolution	In progress	0 Seconds	24 Days 9 Hours	7,312.53%
SCTASK0010018	Employee Onboarding SLA	SLA		In progress	1 Day 23 Hours 57 Minutes	2 Minutes	0.08%
INC0000060	Priority 3 resolution (1 day)	SLA	Resolution	Cancelled	1 Day	0 Seconds	0%
SCTASK0010015	Employee Onboarding SLA	SLA		In progress	1 Day 4 Hours 56 Minutes	19 Hours 3 Minutes	39.72%
INC0000039	Network group resolution	OLA	Resolution	In progress	0 Seconds	91 Days 12 Hours 27 Minutes	54,911.61%
INC0000040	Priority 3 resolution (1 day)	SLA	Resolution	Paused	1 Day	0 Seconds	0%
SCTASK0010017	Employee Onboarding SLA	SLA		In progress	1 Day 12 Hours 36 Minutes	11 Hours 23 Minutes	23.75%
INC0000019	Priority 2 resolution (8 hour)	SLA	Resolution	In progress	0 Seconds	24 Days 9 Hours 14 Minutes	7,315.45%
INC0000060	SAN 001 contract resolution (3.5 hour)	Underpinning contract	Resolution	Breached	0 Seconds	1 Day 38 Minutes	704.1%

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11.Reports



8. Advantages & Disadvantages

Advantages

- Reduces manualeffort.
- Fasteronboardingand offboarding.
- Automatedapprovals.
- Automatic departmentaltask assignment.
- Centralizedemployee lifecycle tracking.
- Improvedtransparency.
- SLAmonitoring.
- Betterreportinganddashboards.
- EnhancedsecuritythroughACLs.

- Scalable workflow automation.

Disadvantages

- Depends on ServiceNow platform availability.
- Initial configuration requires administrative knowledge.
- Additional integrations (HRMS, Active Directory) require further implementation.
- Users require basic training to use the system effectively.

9. Conclusion

The **Automated Employee Onboarding & Offboarding System** successfully simplifies employee lifecycle management by replacing manual processes with an automated ServiceNow workflow. The project demonstrates how ServiceNow's low-code capabilities can improve efficiency, collaboration, security, and transparency. The implemented solution fulfills the functional requirements and provides a scalable foundation for future enterprise HR automation.

10. Future Scope

Future enhancements include:

- Active Directory integration
- HRMS integration
- Employee asset management
- Mobile approvals
- AI-based request recommendations
- Integration with payroll systems • Advanced analytics dashboards
- Multi-level approval workflows
- Integration Hub support
- Automated user provisioning

GitHub Repository

<https://github.com/Sowmya1321/Employee-Lifecycle-Management-ServiceNow>

Project Demo Video

<https://drive.google.com/.....>